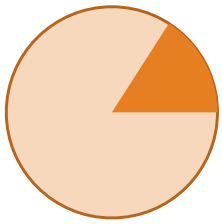
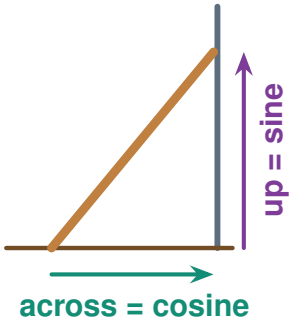
The Math Toolkit

BIG IDEA

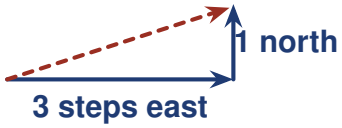
A few simple math tools let us turn flashes of light into a map.



one slice = 1 radian
(crust as long as the radius)



an arrow says
how far and which way



THE IDEA

Big numbers get a shortcut: 10^3 means 1 followed by 3 zeros, which is 1000. An angle can be measured in **radians**: a pizza slice whose crust is as long as the cut from the middle is one radian. Lean a stick on a wall: how far the bottom is from the wall is the **cosine**, and how high the top reaches is the **sine**.

REMEMBER

- ✓ An arrow (a **vector**) tells how far and which way.
- ✓ The **average** tells where things landed. The **spread** tells how scattered they are.
- ✓ A good measurement has an average right on target *and* a small spread.

TRY THIS

Throw ten crumpled paper balls at a target on the floor. Mark where they land. Where is the middle of the group? How spread out are they?

The full module is in the Reference Book.